GGG-T-546

JANUARY 7, 1955

SUPERSEDING
Int. Fed. Spec. GGG–T–00546 (Navy–Ships)
10 November 1952

FEDERAL SPECIFICATION

TONGS, BLACKSMITHS'

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers tongs used primarily for handling material heated in forges and furnaces.

1.1.1 *Federal specification coverage.*—This specification does not cover all types of blacksmiths' tongs, but only those most generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types.*—Blacksmiths' tongs shall be of the following types, as specified (see 6.1).

Type I.—Straight lip, flat.

Type II.—Curved lip, fluted jaw.

Type III.—Rivet, holding.

Type IV.—Single pickup.

Type V.—Double pickup.

Type VI.—Gad.

Type VII.—"V" shaped.

1.2.2 *Sizes.*—Sizes shall be as shown in the table for each type, as specified.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specification:

QQ–S–633—Steel, Carbon; Bars (General Purpose).

Military Specification:

MIL–P–15424—Packaging of Hand Tools for Domestic and Overseas Shipment and Storage.

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

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3. REQUIREMENTS

3.1 Material.—Blacksmiths' tongs, covered by this specification, shall be of carbon steel in the range of FS 1020 to 1050 in accordance with Federal Specification QQ–S–633.

3.1.1 *Illustrations.*—The illustrations shown herein are descriptive and not restrictive, and are not intended to preclude the purchase of blacksmiths' tongs otherwise conforming to this specification.

3.2 Design.—The blacksmiths' tongs shall be similar in design to the figures to which reference is made herein. The two halves of the tongs shall be held together by means of a pin or rivet headed on both ends which shall permit the tongs to be easily operated.

3.2.1 *Handles.*—The handles shall be integral with the body of the tongs or securely welded thereto and may be tapered or straight, at the option of the manufacturer. The handles shall be smooth and of such shape as to be comfortable to the hand. The size of handles shall be well proportioned to the work-holding capacity of the tong jaws.

3.2.2 *Lap joints.*—All blacksmiths' tongs shall have lap joints and the mating surfaces shall be flat and smooth. The holes of both halves shall be finished by a suitable method which produces accurate openings for the rivet or pin. The holes shall be not more than 0.010 inch larger than the rivet or pin.

3.3 Identification marking.—All blacksmiths' tongs shall be marked in a permanent and legible manner with the manufacturer's name or with a trade-mark of such known character that the source of manufacture may be readily determined.

3.4 Finish.—All blacksmiths' tongs shall have either a painted or a natural finish and be free of scale or rust.

3.5 Type I, straight-lip, flat tongs.—Type I tongs shall be suitable for holding flat work, may have a small triangular groove in the jaws, and the tong handles shall close to a satisfactory one-hand grip when clamping stock of the size indicated in table I. They shall conform to table I for the size specified and shall be similar to figure 1.

TABLE I.—*Type I, straight-lip, flat, blacksmiths' tongs*

Size, over all length ±1/8	Distance between jaws when in parallel position	Width of jaws, +5/16, −1/16	Average thickness of each jaw, +3/16, −1/16	Distance between rivet center and end of jaws, +3/8, −1/8	Average diameter of handles, ±1/16	Weight, +1/2, −1/4
Inches	*Inches*	*Inch*	*Inch*	*Inches*	*Inch*	*Pounds*
16	0 to 1/16	3/4	5/16	3 1/8	7/16	2 1/4
18	7/32 to 9/32	7/8	5/8	3 1/8	1/2	2 1/2
20	15/32 to 17/32	7/8	5/8	3 1/8	1/2	2 3/4
22	23/32 to 25/32	1	5/8	3 1/8	9/16	3
24	31/32 to 1 1/32	1	5/8	3 1/8	9/16	3 1/4

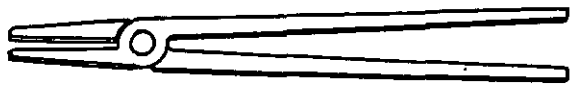


FIGURE 1.—Type I, straight-lip, flat, blacksmiths' tongs.

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3.6 Type II, curved-lip, fluted-jaw tongs.—Type II tongs shall be suitable for holding bolts and other round work. The tong handles shall close to a satisfactory one-hand grip when clamping stock of the size indicated in table II. They shall conform to table II for the size specified and be similar to figure 2.

TABLE II.—*Type II, curved-lip, fluted-jaw, blacksmith' tongs*

Size, inside diameter of fluted jaws, ±10 percent	Over-all length ±½	Average thickness of each fluted jaw +⅛, -1/16	Inside diameter of lips +½, -¼	Outside diameter of lips +½, -¼	Distance between rivet center and end of jaws +1, -½	Average diameter of handles ±1/16	Weight +½, -¼
Inches	*Inches*	*Inch*	*Inches*	*Inches*	*Inches*	*Inch*	*Pounds*
¼	18	5/16	1½	2	3½	7/16	2⅝
⅜	18	5/16	1½	2	3½	7/16	2⅝
½	20	5/16	1⅝	2¼	3½	½	2¾
⅝	22	5/16	1⅝	2⅝	3½	½	3
¾	22	5/16	1⅝	2⅝	3½	½	3
⅞	24	5/16	1¾	2½	3¾	½	3¼
1	24	5/16	1¾	2½	3¾	½	3¼
1½	26	5/16	2	3	4½	½	3⅝

NOTE.—The diameters shall be measured when the tongs are gripping the size of stock indicated in column 1.



FIGURE 2.—Type II, curved-lip, fluted jaw, blacksmiths' tongs.

3.7 Type III, rivet-holding tongs.—Type III tongs shall be suitable for handling hot rivets. They shall conform to table III for the size specified and shall be similar to figure 3.

TABLE III.—*Type III, rivet-holding blacksmiths' tongs*

Size, over-all length ±½	Inside diameter of jaws (closed) ±⅛	Outside diameter of jaws (closed) ±⅛	Average width of jaws ±⅛	Distance between rivet center and end of jaws ±½	Average diameter of handles ±1/16	Weight ±⅝
Inches	*Inch*	*Inches*	*Inch*	*Inches*	*Inch*	*Pounds*
18	⅛	⅛	7/16	4½	7/16	2⅛
24	⅝	1⅛	⅝	4½	7/16	2⅛

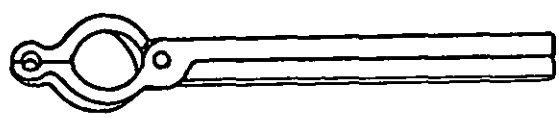


FIGURE 3.—Type III, rivet-holding, blacksmiths' tongs.

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3.8 Type IV, single-pickup tongs.—Type IV tongs shall be suitable for handling medium-heavy flat work. They shall conform to table IV for the size specified and shall be similar to figure 4.

TABLE IV.—*Type IV, single-pickup, blacksmiths' tongs*

Size, over-all length, ±½	Average width of jaws, +⅝, −⅛	Average thickness of each jaw extreme end, ±1/16	Average thickness of each jaw at the lip diameter, ±⅛	Distance between rivet center and end of jaws, +¾, −½	Average diameter of handles, ±1/16	Weight, +¾, −½
Inches	*Inch*	*Inch*	*Inch*	*Inches*	*Inch*	*Pounds*
18	⅝	5/16	7/16	4	½	2
20	⅝	5/16	7/16	4	½	2⅛
24	¾	5/16	7/16	4	½	2⅜

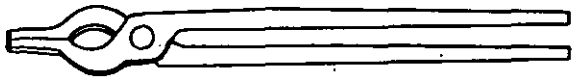


FIGURE 4.—Type IV, single-pickup, blacksmiths' tongs.

3.9 Type V, double-pickup tongs.—Type V tongs shall be suitable for handling flat and round work. They shall conform to table V for the size specified and shall be similar to figure 5.

TABLE V.—*Type V, double-pickup, blacksmiths' tongs*

Size, over-all length, ±½	Average width of jaws, +¼, −⅛	Average thickness of each jaw at extreme end, ±1/16	Average thickness of each jaw at the front lip diameter, ±3/32	Average thickness of each jaw at the rear lip diameter, +3/16, −⅛	Distance between rivet center and end of jaws, ±1	Average diameter of handles, ±1/16	Weight, ±½
Inches	*Inch*	*Inch*	*Inch*	*Inch*	*Inches*	*Inch*	*Pounds*
18	½	7/32	9/32	7/16	4¾	7/16	2
20	½	7/32	9/32	7/16	4¾	7/16	2⅛
24	½	7/32	9/32	7/16	4¾	7/16	2⅜

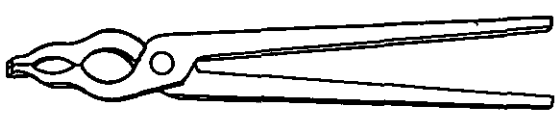


FIGURE 5.—Type V, double-pickup, blacksmiths' tongs.

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3.10 Type VI, gad tongs.—Type VI tongs shall be suitable for general purpose forging. They shall conform to table VI for the size specified and shall be similar to figure 6.

TABLE VI.—*Type VI, gad, blacksmiths' tongs*

Size, over-all length, +½, -½	Average width of jaws, +3/16, -⅛	Average thickness of each jaw at the extreme end, +⅛, -⅛	Average thickness of each jaw at the lip diameter, +3/16, -1/16	Distance between rivet center and end of jaws, ±1	Average diameter of handles, +1/16, -1/16	Weight, +½, -½
Inches	*Inch*	*Inch*	*Inch*	*Inches*	*Inch*	*Pounds*
18	¾	9/32	⅜	4	7/16	2½
20	⅞	5/16	⅜	4¼	½	3
24	⅞	5/16	⅜	4¼	½	3¼



FIGURE 6.—Type VI, gad, blacksmiths' tongs.

3.11 Type VII, "V" shaped tongs.—Type VII tongs shall have a V groove running lengthwise on the inside of each jaw and be suitable for holding square and round stock. The tong handles shall close to a satisfactory one-hand grip when clamping stock of the size specified. They shall conform to table VII for the size specified and shall be similar to figure 7.

TABLE VII.—*Type VII, "V" shaped, blacksmiths' tongs*

Size, jaw opening or square stock size (nominal)	Over-all length, +½, -½	Distance between rivet center and end of jaws, +1¾, -⅝	Average diameter of handles, +1/16, -1/16	Weight	
				Min.	Max.
Inch	*Inches*	*Inches*	*Inch*	*Pounds*	*Pounds*
¼	18	3⅝	7/16	2⅛	2⅝
⅜	20	3⅝	7/16	2½	3
½	20	3⅝	½	2¾	3¼
⅝	22	3⅝	½	2⅞	3⅜
¾	22	3⅝	½	3	3½
⅞	24	3⅝	½	3⅛	3¾
1	24	3⅝	½	3⅛	4½

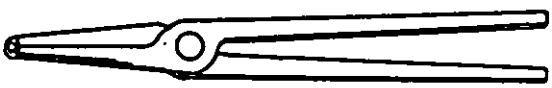


FIGURE 7.—Type VII, "V" shaped, blacksmiths' tongs.

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3.12 Workmanship.—The workmanship shall be first class in every respect, and all tongs shall be free from defects which may adversely affect their appearance, serviceability, or operation.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling for lot acceptance.

4.1.1 *Inspection lot.*—All blacksmiths' tongs of the same size and type presented at one time shall be considered a lot for purposes of acceptance inspection and tests.

4.1.2 *Sampling for inspection.*—A random sample of tongs shall be taken as specified in table VIII from each inspection lot of material offered for Government inspection of visual and dimensional characteristics with lot acceptance based on the inspection requirements in accordance with Military Standard MIL–STD–105.

TABLE VIII.—*Sampling for visual and dimensional inspection AQL (approx.) = 1.5 percent defective*

Number of tongs in an inspection lot	Sample size, number of tongs to be inspected (See Military Standard MIL–STD–105)
40 and under	Inspection level III
41 to 300	Inspection level II
301 and over	Inspection level I

4.2 Inspection.

4.2.1 *Visual, dimensional, and operational inspection.*—Each of the sample tongs taken in accordance with table VIII shall be visually and dimensionally inspected and subjected to a practical service operation at the place of manufacture by the Government inspector to verify compliance with this specification. Any tong in the sample containing one or more visual, dimensional, or operational defects shall be rejected, and if the number of defective tongs in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3 Rejected lots.—Rejected lots may be offered again for Government inspection provided the contractor has repaired or removed all nonconforming tongs. The Government inspector shall again select and examine samples from such resubmitted lots to verify compliance with this specification.

5. PREPARATION FOR DELIVERY

5.1 For domestic shipment, immediate use.

5.1.1 *Preservation.*—Cleaning and preservation shall be in accordance with the manufacturer's standard practice.

5.1.2 *Packaging.*—Commercial packages will be acceptable.

5.1.3 *Packing.*—Unless otherwise specified, the subject commodity shall be prepared for shipment to permit acceptance by carrier for transportation at lowest applicable rate and to afford maximum protection from normal hazards of transportation.

5.1.4 *Marking.*

5.1.4.1 *Interior containers.*—Unless otherwise specified, packages and interior containers shall be marked with the name of the item, type and size, and the name or trade-mark of the contractor of such known character that the source of supply may be readily determined.

5.1.4.2 *Exterior containers.*—Unless otherwise specified, shipping containers shall be marked with the name of the item, the type, the size, and the quantity contained therein, as defined by the contract or order under which shipment is made, the name of the contractor, and the number of the contract or order.

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5.1.4.3 *For Military purchases.*—In addition to any special marking required by the contract or order, interior and exterior containers shall be marked in accordance with Military Standard MIL–STD–129.

5.2 For domestic shipment and storage or overseas shipment.—Blacksmiths' tongs shall be cleaned, preserved, packaged, packed, and marked in accordance with Military Specification MIL–P–15424.

6. NOTES

6.1 Ordering data.—Procurement documents should specify the following and purchasers should exercise any desired options offered herein.

(a) Title, number, and date of this specification.

(b) Type and size of blacksmiths' tongs required. (See 1.2.1, 1.2.2, and applicable table.)

(c) Whether for domestic shipment immediate use, domestic shipment and storage, or overseas shipment. (See 5.1 and 5.2.)

6.2 THE ILLUSTRATIONS HEREIN ARE DESCRIPTIVE AND NOT RESTRICTIVE, and are not intended to preclude the purchase of blacksmiths' tongs otherwise conforming to this specification.

6.3 This specification *conforms* with Simplified Practice Recommendations R17–47 insofar as practicable.

6.4 Transportation description.

6.4.1 The transportation description applicable to these items is:

Blacksmiths' anvil tools:
Carload weight 36,000 pounds.
Truckload weight 36,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:

Army—O E T

Navy—Sh MC Or Y

Air Force.

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